

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP T, S, DO, pH	WQP 018	1.0	

PREPARED BY			
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1.0	1/29/24	Initial Release for using the Eureka Manta+ 20 sonde	TAC

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1. Title

- 1.1.** Standard Operating Procedure (SOP) for seawater Depth, Temperature, Salinity, Dissolved Oxygen, and pH determination.

2. Applicability

- 2.1.** This SOP applies to the determination of seawater Depth, Temperature, Salinity, Dissolved Oxygen, and pH profiles at shallow (<30 m) coral reef sites around Puerto Rico.

3. Purpose

- 3.1.** Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze seawater Depth, Temperature, Salinity, Dissolved Oxygen, and pH profiles at shallow coral reef sites around Puerto Rico.
- 3.2.** Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater Depth, Temperature, Salinity, Dissolved Oxygen, and pH data following the Quality Assurance Project Plan (QAPP).

4. Methods

- 4.1.** Standard Method 2550 B (Baird et al., 2017) for Temperature.
- 4.2.** Standard Method 2520 B (Baird et al., 2017) for Salinity.
- 4.3.** Standard Method 4500-O/H (Baird et al., 2017) for Dissolved Oxygen.
- 4.4.** EPA Method 150.2 (Environmental Protection Agency) for pH.
- 4.5.** Eureka Manta+ 20 Sonde is used to measure *in situ* seawater surface and bottom Depth, Temperature, Salinity, Dissolved Oxygen, and pH profiles.

5. Health and Safety

- 5.1.** Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2.** When loosening removable parts of a manifold probe, face the probe away from your body and others, as pressure may build up within the instrument resulting in caps, sensors, or other parts that can shoot out and cause injury.

6. General Precautions

- 6.1.** Always maintain two points of contact when deploying the Eureka sonde on the boat to avoid losing the instrument overboard.
- 6.2.** Ensure the boat engines are in idle and the captain approves of the sonde being deployed overboard before deploying the sonde to ensure that the instrument cables do not become entangled with the boat propeller.
- 6.3.** If there are problems with the sonde or software, call the manufacturer (512-302-4333) or email info@waterprobes.com.
- 6.4.** Do not allow the sensors on the sonde to dry.
- 6.5.** For storage after deployment:

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- 6.5.1. Rinse the bluetooth transmitter, the sonde, sonde's sensors, and the sensor guard with tap water
- 6.5.2. Rinse the storage cup with tap water
- 6.5.3. Remove the sensor guard and replace it with the storage cup while it holds a few ounces of tap water to allow the sensors to remain humid but not submerged.
 - 6.5.3.1. Sensors should never be left fully submerged or allowed to dry.

7. Training

- 7.1.** Personnel shall be trained to calibrate, run calibration checks, and operate the Eureka Manta+ 20 Sonde before use.
- 7.2.** Personnel shall receive training on how to complete the corresponding forms for this SOP.

8. Required Materials

- 8.1.** Eureka Manta+ 20 Multiparameter Sonde
- 8.2.** Eureka Manta Mobile Bluetooth emitter
 - 8.2.1. Eureka Bluetooth Emitter Smart Charger
 - 8.2.2. Eureka Bluetooth Emitter Smart Charger AC cable
- 8.3.** Field deployment cable
- 8.4.** Laptop computer or tablet (PC)
 - 8.4.1. *Manta+ 20 Control Software* or *MantaLink* installed
- 8.5.** Laptop computer or tablet (PC) chargers
 - 8.5.1. Dell charger
 - 8.5.2. iPad charger
- 8.6.** NIST Certified Thermometer
- 8.7.** Mild detergent
- 8.8.** Warm water
- 8.9.** Soft toothbrush
- 8.10.** Deionized water
- 8.11.** pH standards (7.0, 9.0, and 10.0 or similar ranged standards)
- 8.12.** Conductivity standards (50 mS/cm and 100mS/cm or similar ranged standards)
- 8.13.** Air saturated water
- 8.14.** Eureka Sonde Daily Check (Appendix J)
- 8.15.** Field Form (Appendix F)

9. Calibration Checks

- 9.1.** Prior to each field sampling day, perform a calibration check for each parameter.
 - 9.1.1. Eureka Manta+ 20 sensors should be checked for calibration in the following order:
 - 9.1.1.1. Depth
 - 9.1.1.2. Temperature
 - 9.1.1.3. Conductivity
 - 9.1.1.4. pH
 - 9.1.1.5. LDO
 - 9.1.2. Since there is no *Calibration Check* function in the Eureka Manta+ 20 software, calibration checks only require that real-time measurements using known

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standards fall within the expected accuracy/precision values for the Eureka Manta+ 20:

Parameter	Accuracy
Depth	±0.4 m
Temperature	±0.1°C
Salinity (as Specific Conductance)	±0.5% (± 0.25 mS/cm for a 50 mS/cm std)
pH	±0.1
Oxygen	± 0.1 mg/L or ± 1.5 %

9.2. Before starting a calibration check:

- 9.2.1. Rinse at least 2 times with deionized water by filling the storage cup, screwing the cap on, shaking to touch all surfaces, dumping the water, and repeating.
- 9.2.2. Make sure all measurements are done with the sensors facing up towards the ceiling.

9.3. To perform a calibration check:

- 9.3.1. Turn on your PC
 - 9.3.1.1. Laptop password: Colegio12345
 - 9.3.1.2. iPad password: 123456
- 9.3.2. Ensure the Manta Mobile Bluetooth emitter is securely fastened to the Eureka Manta sonde cable using the screws.
- 9.3.3. Turn on the Manta Mobile Bluetooth emitter.
- 9.3.4. If no previous bluetooth connection has been established prior to calibration, refer to page 53 of the Eureka MANTA and TRIMETER MULTIPROBE MANUAL.
 - 9.3.4.1. If you have established a bluetooth connection, but the PC still fails to recognize the Manta+ 20 Sonde:
 - 9.3.4.1.1. Turn ON/OFF your bluetooth or go to you PC's *Settings* → *Privacy & Security* → *Bluetooth* → turn ON/OFF accessibility to MantaLink.
- 9.3.5. Once a bluetooth connection is established between the Manta+ 20 Sonde and Bluetooth Emitter you should be able to see a display window showing real-time measurements of whatever is in contact with the probe's sensors.
 - 9.3.5.1. You are now ready to begin calibration checks.

9.4. Depth calibration check

- 9.4.1. Make sure the Manta is not in the water
- 9.4.2. Make sure the pressure port is not clogged.
 - 9.4.2.1. If it is, clear the obstruction using something soft or a toothpick.

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- 9.4.3. Complete steps 9.2
- 9.4.4. Place the Manta+ 20 sonde on a stable surface
 - 9.4.4.1. Remove storage cup and use kimwipes or a soft microfiber cloth to dry out the depth sensor
- 9.4.5. While on the MantaLink *Home* screen, depth should be reading $0.0 \text{ m} \pm 0.4 \text{ m}$.
 - 9.4.5.1. If outside of calibration range, re-calibrate sonde following step 10.3.
- 9.4.6. Record observations in Appendix J Eureka Sonde Daily Check.

9.5. Temperature calibration check

- 9.5.1. Complete steps 9.2
- 9.5.2. The temperature sensor can only be calibrated by the manufacturer.
- 9.5.3. Instead of calibration, perform a calibration check using a NIST certified thermometer prior to each sampling event to ensure accuracy and control for any potential sensor drift throughout the sampling event.
- 9.5.4. Place the Manta+ 20 Sonde near a sink to prepare for calibration check.
- 9.5.5. While on the MantaLink *Home* screen (showing real-time data), fill the calibration cup with tap water at room temperature.
- 9.5.6. Place the NIST certified thermometer inside the calibration cup with room-temperature water so that the tip is suspended in the middle of the water in the calibration cup and allow the temperature measurements to stabilize.
 - 9.5.6.1. If the recorded temperatures are within $\pm 0.1^\circ\text{C}$ from each other, the temperature sensor is correctly calibrated.
 - 9.5.6.2. Record observations in Appendix J.
- 9.5.7. If the recorded temperatures **are NOT** within $\pm 0.1^\circ\text{C}$ (<20%) from each other, contact Dr. Travis Courtney or customer support as factory service might be needed.
 - 9.5.7.1. **Note:** Deviations in the thermistor are very uncommon and are more likely to be the result of improper calibration check procedure than drift of the thermistor.

9.6. Conductivity calibration check

- 9.6.1. Complete steps 9.2
- 9.6.2. Rinse the storage cup with 50 mS/cm conductivity standard by adding a few ounces to the storage cup, screwing it closed, and shaking vigorously for 6 seconds.
 - 9.6.2.1. Repeat for a total of two times.
- 9.6.3. While on the *Home* screen and with the sondes sensors facing upwards, fill the storage cup with the 50 mS/cm standard solution until you cover the sensors for $\sim 1 \text{ cm}$.
- 9.6.4. Allow the sensors to fully stabilize.
- 9.6.5. Conductivity should be $50 \pm 0.25 \text{ mS/cm}$ (or 0.5%) for a 50 mS/cm standard.
 - 9.6.5.1. Record observations in Appendix J Eureka Sonde Daily Check.

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9.6.5.2. Note, the 50 mS/cm standard may have its own reported uncertainty range, which may be greater than the precision of the Eureka Sonde. In that case, the calibration is only as good as the standard and the conductivity measurement should be within the reported uncertainty of the standard.

9.6.6. If outside of the calibration range (± 0.25 mS/cm; 0.5%), re-calibrate sonde following step 10.5.

9.7. pH calibration check

9.7.1. Complete steps 9.2

9.7.2. Rinse the sensors and storage cup by adding a few ounces of pH 9.0 to the storage cup, screwing it closed, and shaking vigorously for 6 seconds.

9.7.2.1. Repeat for a total of two times.

9.7.3. While on the *Home* screen and with the sondes sensors facing upwards, fill the storage cup with the pH 9.0 standard until you cover the sensors for ~ 1 cm.

9.7.4. Allow at least 1 minute for sensor stabilization.

9.7.5. If the recorded pH is within ± 0.1 from the standard, your sonde's pH sensor is calibrated.

9.7.5.1. Record observations in Appendix J Eureka Sonde Daily Check.

9.7.6. If outside of the calibration range (± 0.1), re-calibrate the sonde following step 10.7.

9.8. LDO calibration check

9.8.1. Complete steps 9.2

9.8.2. Fill half of a 1L bottle with deionized water, shake it for 1 minute, open it, and let it stand to let the air bubbles float out.

9.8.3. While on the *Home* screen and with the sonde's sensors facing upwards, fill the storage cup with the air saturated water prepared in the previous step until you cover the sensors by at least ~ 1 cm.

9.8.4. Allow for the sensor to stabilize.

9.8.5. Record observations as %Sat in Appendix J Eureka Sonde Daily Check.

9.8.5.1. DO should be 100 ± 1.5 % saturation.

9.8.6. If outside of the calibration range (± 0.1 mg/L; $\pm 1.5\%$), re-calibrate the sonde following step 10.6.

10. Calibrations

10.1. Eureka Manta+ 20 sensors should be calibrated in the following order:

depth $\rightarrow$ temperature (by manufacturer only) $\rightarrow$ conductivity $\rightarrow$ pH $\rightarrow$ LDO

10.1.1. **Note:** The calibrations are dependent on each other, so for example, LDO requires the sonde to correctly measure depth, temperature, and conductivity to provide an accurate LDO reading.

10.2. To prepare for calibration:

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- 10.2.1. Place Manta+ 20 Sonde in a stable location and close to a sink and to your chemical disposal container.
- 10.2.2. If the sensors are dirty, clean all sensors with warm “soapy” water and a gentle toothbrush. Make sure to **NOT** use anything abrasive and take special care with the LDO sensor and pH probe.
 - 10.2.2.1. After cleaning, rinse thoroughly with tap water.
- 10.2.3. Turn on your iPad.
- 10.2.4. Turn on the Manta Mobile Bluetooth emitter.
- 10.2.5. If no previous bluetooth connection has been established prior to calibration, refer to page 53 of the Eureka MANTA and TRIMETER MULTIPROBE MANUAL.
 - 10.2.5.1. If you have established a bluetooth connection, but the iPad still fails to recognize the Manta+ 20 Sonde:
 - 10.2.5.1.1. Turn ON/OFF your bluetooth or go to you iPad’s *Settings* → *Privacy & Security* → *Bluetooth* → turn ON/OFF accessibility to MantaLink
- 10.2.6. Once a bluetooth connection is established between the Manta+ 20 Sonde and Bluetooth Emitter you should be able to see a display window showing real-time measurements of whatever is in contact with the probe’s sensors.
 - 10.2.6.1. You are now ready to begin calibration.

10.3. Depth calibration

- 10.3.1. Make sure the Manta is not in the water.
- 10.3.2. Make sure the pressure port is not clogged.
 - 10.3.2.1. If it is, clear the obstruction using something soft or a toothpick.
- 10.3.3. While on the MantaLink *Home* screen, *click* on *Calibrate*, select *Depth m*.
- 10.3.4. Follow the instructions on the display window and *click OK* when the reading stabilizes.
 - 10.3.4.1. Fill-out the corresponding information in Calibration Form (Appendix J)
- 10.3.5. *Click OK* if SRF value is between 80-120
- 10.3.6. *Click OK* to confirm calibration.
- 10.3.7. Your sonde’s depth sensor is now calibrated.
 - 10.3.7.1. Follow step 9.4. to run a calibration check.

10.4. Temperature calibration

- 10.4.1. The temperature sensor can only be calibrated by the factory.

10.5. Conductivity calibration

- 10.5.1. While on the MantaLink *Home* screen, *click* on *Calibrate*, select *SpCond mS/cm*.
 - 10.5.1.1. The zero point for the sensor is set electronically, so you need only set the “slope” point.

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- 10.5.2. Make sure the sensors are facing upwards towards the ceiling.
- 10.5.3. Fill ~1/5 of the storage cup with 50 mS/cm standard solution, shake to touch all surfaces in the sensor cup, and discard the solution in the appropriate container.
 - 10.5.3.1. Repeat for a total of two times.
- 10.5.4. Fill the storage cup with 50 mS/cm standard solution until you cover all sensors by at least 1 cm.
 - 10.5.4.1. Tap gently on the cup to make sure there aren't bubbles trapped in the conductivity sensor.
- 10.5.5. Once the value has stabilized, write down the conductivity value in the provided box, and *click OK*.
 - 10.5.5.1. Fill-out the corresponding information in Calibration Form (Appendix J)
- 10.5.6. Click Ok again if the SRF is between 80-120.
- 10.5.7. Your sonde's conductivity sensor is now calibrated.
 - 10.5.7.1. Follow step 9.6. to check the calibration and make sure that conductivity measured by the Manta+ 20 Sonde is within $\pm 0.5\%$ of a known standard.

10.6. pH calibration

- 10.6.1. Make sure that the electrolyte solution has been replaced within one month of the calibration date.
 - 10.6.1.1. If in doubt or > 1 month has elapsed since solution refill:
 - 10.6.1.1.1. Unscrew the cap on the pH sensor using a flat head screwdriver or a coin.
 - 10.6.1.1.2. Discard old electrolyte solution.
 - 10.6.1.1.3. Fill the sleeve completely with the new reference electrolyte solution until it forms a bulge, almost overflowing.
 - 10.6.1.1.4. Tap the sonde to discard any bubbles.
 - 10.6.1.1.5. Screw the cap back into the sleeve. You should notice air and excess electrolyte solution being forced out from the sleeve.
 - 10.6.1.1.6. Rinse sensors using deionized water before proceeding with calibration/check.
- 10.6.2. While on the MantaLink software, click on *Calibrate*, select *pH units*.
- 10.6.3. Fill ~1/5 of the storage cup with pH 7.0 standard solution, shake to touch all surfaces in the sensor cup, and discard the solution in the appropriate container.
 - 10.6.3.1. Repeat for a total of two times.
- 10.6.4. Fill the storage cup with pH 7.0 standard solution until you cover all sensors by at least 1 cm.
 - 10.6.4.1. Tap gently on the cup to make sure there aren't bubbles trapped in the pH sensor.
- 10.6.5. If the Eureka sonde temperature varies from 25.0°C, use the reported temperature value of the pH buffer solution to correct the pH value using the

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conversion table located in the pH standard's bottle and use this temperature corrected value as the certified value for the standard.

10.6.5.1. Once the value has stabilized, write down the pH value in the provided box, and *click OK*.

10.6.5.1.1. Make sure that pH values measured by the Manta+ 20 Sonde are within ± 0.1 of the known standard.

10.6.5.1.2. Fill-out the corresponding information in Calibration Form (Appendix J)

10.6.6. Repeat steps 10.7.3 - 10.7.5. But now using a pH 10.0 solution.

10.6.6.1. Note, alternative pH standard concentrations may be used as long as one is less than and one is greater than the expected pH for seawater of 8.1.

10.6.7. After completing Appendix J, follow step 9.8. to check the calibration and make sure that pH values measured by the Manta+ 20 Sonde are within ± 0.1 of a known standard.

10.7. Dissolved Oxygen calibration

10.7.1. Fill half of a 1 L bottle with deionized water, shake it for 1 minute, open it and let it stand for ~30 seconds to let any air bubbles float out.

10.7.1.1. Note any size or type of half-filled large, clean bottle may be used.

10.7.2. While on the MantaLink *Home* screen, click on *Calibrate*, select *HDO %Sat*.

10.7.3. Fill the storage cup with the solution from step 9.6.1. until you cover all sensors by at least 1 cm.

10.7.3.1. Tap gently on the cup to make sure there aren't bubbles trapped in the sensors

10.7.3.2. Allow for temperature to stabilize.

10.7.4. Record the local daily atmospheric pressure in mm Hg.

10.7.4.1. Go to <https://forecast.weather.gov/>

10.7.4.2. Enter the current location, for example La Parguera and click Go.

10.7.4.3. The current conditions will be reported for the nearest observation station.

10.7.4.4. The current barometer reading will report in units of inches of mercury.

10.7.4.5. Use an online unit converter or multiply the inches of mercury value by 25.4 to convert to mm Hg.

10.7.4.6. Type the current atmospheric pressure in mm Hg into the correct box and click OK to proceed to the next step.

10.7.4.7. Fill-out the atmospheric pressure in Calibration Form (Appendix J)

10.7.5. Once the value has stabilized, write down the *HDO %Sat* value in the provided box, and *click OK*.

10.7.5.1. Fill-out the corresponding information in Calibration Form (Appendix J)

10.7.6. *Click Ok* again if the SRF is between 80-120

10.7.7. Your sonde's dissolved oxygen sensor is now calibrated.

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- 10.7.7.1. Follow step 9.7. to check the calibration and make sure that dissolved oxygen measured by the Manta+ 20 Sonde is $\pm 1.5\%$ of 100% saturation.

11. Preparing for field deployment and sample collection

- 11.1. **Optional:** Prior to each quarterly sampling event, the name of the AutoSnap log file can be reset to the corresponding quarter to streamline data organization.
 - 11.1.1. The AutoSnap records the date and time of sampling within the log so all data is preserved in an indexable format regardless of the file name.
 - 11.1.2. Note, the iPad also has the capacity to record latitude and longitude using the internal GPS although Out of Range issues prevented us from relying on this feature.
- 11.2. Make sure that the iPad and Bluetooth emitter are fully charged.
- 11.3. Complete step 9 (mandatory) and step 10 when necessary.
- 11.4. Turn on your iPad.
- 11.5. Turn on the Manta Mobile Bluetooth emitter.
- 11.6. If no previous bluetooth connection has been established prior to calibration, refer to page 53 of the Eureka MANTA and TRIMETER MULTIPROBE MANUAL.
 - 11.6.1. If you have established a bluetooth connection, but the iPad still fails to recognize the Manta+ 20 Sonde:
 - 11.6.1.1. Turn ON/OFF your bluetooth or go to you iPad's *Settings* → *Privacy & Security* → *Bluetooth* → turn ON/OFF accessibility to MantaLink
- 11.7. Once a Bluetooth connection is established between the Manta+ 20 Sonde and Bluetooth Emitter you should be able to see a display window showing real-time measurements of whatever is in contact with the probe's sensors.
- 11.8. Once at the sampling site, replace the storage cup with the sensor guard for deployment.
 - 11.8.1. If the storage/calibration cup is not available, keep the sensors covered with the sensor guard inside a bucket of water.
- 11.9. While a connection is established between the Bluetooth emitter and your PC, select *AutoSnap* to begin logging.
- 11.10. After receiving the captain's approval, gently lower the Manta+ 20 Sonde ~0.5 m into the water.
 - 11.10.1. Allow the sensors to stabilize.
 - 11.10.2. Fill-out the corresponding information on the Sampling Log (Appendix F).
- 11.11. Lower the sonde at a slow and steady rate until it gets ~ 0.5m from the bottom with caution not to touch the bottom, stir sediment up, or entangle the sonde or deployment cable.
 - 11.11.1. Allow the sensors to stabilize.
 - 11.11.2. Fill-out the corresponding information on the Sampling Log (Appendix F).
- 11.12. Gently pull the sonde back up into the boat and place it in a bucket with tap or sea water while still having the sensor guard on.
- 11.13. Stop the *AutoSnap* function by tapping the *AutoSnap* icon again and place the iPad to

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sleep until the next site.

11.14. Turn off the Manta Mobile Bluetooth emitter to preserve battery.

11.15. After sampling the last site, rinse all equipment and data sheets with tap water and let air dry avoiding direct sunlight.

11.15.1. Rinse the Eureka sonde, sensors, Bluetooth Emitter, iPad, and deployment cable with tap water.

11.15.1.1. Rinse and replace the sensor guard with the storage cup while it holds a few ounces of tap water.

12. Data transfer

12.1. Turn on your iPad and make sure it is connected to the internet.

12.2. Turn on the Manta Mobile Bluetooth emitter.

12.3. If no previous bluetooth connection has been established prior to calibration, refer to page 53 of the Eureka MANTA and TRIMETER MULTIPROBE MANUAL.

12.3.1. If you have established a bluetooth connection, but the iPad still fails to recognize the Manta+ 20 Sonde:

12.3.1.1. Turn ON/OFF your bluetooth or go to you iPad's *Settings* → *Privacy & Security* → *Bluetooth* → turn ON/OFF accessibility to MantaLink

12.3.2. Once a Bluetooth connection is established between the Manta+ 20 Sonde and Bluetooth Emitter you should be able to see a display window showing real-time measurements of whatever is in contact with the probe's sensors.

12.4. To download *Snapshots* click on *Files* → while on the *Snapshot* tab, click the snapshot file you wish to download → a table showing data for each snapshot should pop on the right half of the iPad screen

12.4.1. Using this table, confirm the dates and times of the files that you wish to download

12.4.2. Click on the share button  located to the right of the "Snapshot Files" title and select the *snapshot(s)* you wish to download

12.4.3. Click "Email" to receive a .csv file with the *snapshots* to an email address you select.

12.5. Send the email with the .csv file and proceed to backup the data on the CCRI laptop and Google Drive.

13. Documentation

13.1. Field data will be recorded in the Sampling Log (Appendix F).

13.2. Calibration data will be recorded in the Calibration Form (Appendix J).

13.3. Field and Calibration forms shall be kept in the designated file cabinet at CCRI water quality lab.

13.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.

13.5. All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.

13.5.1. All data will be electronically archived and backed up online.

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14. Cleaning and Maintenance

- 14.1.** After each sampling event, rinse the field iPad, sonde, sonde's sensors, storage cup, and the sensor guard thoroughly with clean warm fresh water.
- 14.2.** After each sampling event, inspect sensors carefully for physical damage.
 - 14.2.1.** In case of unwanted material deposits over the sensors, gently clean sensors with a soft brush and mild detergent.
- 14.3.** Fully charge laptop, iPad, and Bluetooth emitter.
- 14.4.** For long-term storage (>24hr), rinse the storage cup and sensors with tap water and screw the storage cup on the sensors while it is still humid.
 - 14.4.1.** Keep a few ounces of tap water in the storage/calibration cup when the Manta is not being used.
 - 14.4.2.** Do not keep the storage cup with distilled water for long periods of time as this may damage the probes.
- 14.5.** Perform preventive maintenance prior to each quarterly sampling cycle.
 - 14.5.1.** The pH electrolyte solution should be changed within a month of each pH calibration to ensure proper functioning of the electrode.
 - 14.5.2.** Carefully inspect sensors, O-rings, connective pins, and physical integrity of the sonde in general.
 - 14.5.3.** Refer to the Eureka MANTA and TRIMETER MULTIPROBE MANUAL for instructions.
 - 14.5.4.** Record maintenance steps in the instrument maintenance log.
- 14.6.** If a factory service is needed, perform a calibration check or calibration (if needed) once the sonde is received to make sure it is within working standards.

15. Troubleshooting

- 15.1.** In case of reading errors, check the sensor path for bubbles, grease, or other contamination.
- 15.2.** In case of continuous unstable readings, contact the manufacturer to evaluate replacing the sensors.
- 15.3.** In case precision for any sensor's calibration check is not met, a calibration of that specific sensor should be performed.

16. QA/QC

- 16.1.** Eureka recommends calibration when measurements are outside of established QA/QC thresholds for the respective parameter.
- 16.2.** Temperature (± 0.1 °C), depth (± 0.4 m), conductivity ($\pm 0.5\%$), dissolved oxygen ($\pm 1.5\%$), and pH (± 0.1) sensors will be checked for calibration using the NIST certified thermometer and/or standard solutions prior to any field sampling event.
 - 16.2.1.** In case precision for each parameter's calibration check is not met, then a calibration of each parameter will occur.
 - 16.2.2.** Temperature can only be calibrated by the manufacturer.
- 16.3.** Additionally, all field samples should adhere to typical seasonal values observed in coastal waters of Puerto Rico.
 - 16.3.1.** Any seemingly erroneous measurements will be flagged for further review in

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accordance with the QAPP.

17. References

- 17.1. Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.
- 17.2. Eureka Water Probes (rev. 11-2018) MANTA and TRIMETER MULTIPROBE MANUAL. <https://s.campbellsci.com/documents/ca/manuals/Manta.pdf>